Chinmay Goyal

Senior Year Undergraduate — Computer Science and Engineering — IIT Kanpur
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Education

Year	Degree	Institute	CPI/Percent
2018 - Present	B.Tech., Computer Science and Engineering (Grad. 2022)	Indian Institute of Technology Kanpur	9.8/10.0
2018	Class XII (CBSE - AISSCE)	Rukmani Birla Modern High School, Jaipur	95.2%
2016	Class X (CBSE - AISSE)	Rukmani Birla Modern High School, Jaipur	10.0/10.0

Achievements

- Received **Academic Excellence Award** for the years 2018-19 and 2019-20 at IIT Kanpur
- Secured an All India Rank **166**, among 0.23 million candidates in **JEE Advanced** 2018, organized by IIT Kanpur
- Achieved an All India Rank **817**, among 1.2 million candidates in **JEE Main** 2018, conducted by CBSE
- Attained All India Rank **153** and selected for **Kishore Vaigyanik Protsahan Yojana**(KVPY) fellowship in 2016-17
- Secured **National Talent Search Examination, 2016** (NTSE) scholarship and **State Rank 1** (Stage-1)
- Among **National Top 1%** in **National Standard Examination in Physics**, 2017 organised by IAPT
- Represented IIT Kanpur at the **ICPC 2020 Asia Amritapuri Regional Contest** among 700+ selected teams

Internship Experience

TOWER RESEARCH CAPITAL — Project: **An Aggressive Model for Trading** [May '21 - July '21]
- Objective: Enforce strategies to generate an **aggressive trading model** to execute profitable trades based on market behaviour
- Constructed model to predict the market behaviour of a product using **classification** and **regression** techniques
- Compared performance of different models through variations in **Precision-Recall Curves**, **F Scores** and **Profit-Loss Ratio**
- Achieved an increment of profit by **150%** in comparison to a simple Linear Regression Model

SUPERGAMING — Project: **Bull Fight** - An online multiplayer strategy based game [April '20 - Jun '20]
- Objective: Creating new points of interaction and content for the players to increase user retention
- Included new **gameplay elements** and implemented a new **First Time User Experience** through **Unity** game engine
- Improved the **core game mechanics** through **C#** scripting and provided new variables that affect the outcome of a game
- Engineered Deck Making mechanism to allow players to customize their deck as per need

Projects Undertaken

C-- : C Compiler — Mentor: Prof. Amey Karkare, CSE, IIT Kanpur — Compiler Design [Jan '21 - Apr '21]
- Implemented a **compiler** for **C** programming language with the assembly language generated for **x86_32 bit architecture**
- Used Python library, **PLY** to implement the **lexer** to tokenize the input and **parser** to assign to correct grammatical rule
- Used **3 Address Code** as **intermediate language** to act as a bridge between high-level source & low-level assembly language
- Added support for errors and warnings generation for syntactically and semantically erroneous program

GEM OS — Mentor: Prof. Debadatta Mishra, CSE, IIT Kanpur — Operating System [Aug '20 - Nov '20]
- Implemented various operating system elements on GemOS such as **Paging**, **File system** and **Message Queues**
- Engineered memory management operations with provisions for **mmap**, **munmap** and **huge page creation**
- Created a **debugging system** for GemOS that could dump register information at the start of any function.

RESEARCH PORTAL — Mentor: Prof. Sandeep Shukla — Academics and Career Council [May '19 - July '19]
- Spearheaded the creation of an online portal to bridge the gap between students and faculty members
- The portal acts as a platform for research related interaction and project collaboration
- Used **Node** for back end, **MongoDB** as database and designed user interface with **HTML5** and **CSS**

C3i LAB, IIT Kanpur — Project: Filtering USB Storage Devices in Linux Based Systems [Dec '19]
- Worked on methods to **allow selective USB Devices** to transfer system data using their unique **Product ID**
- Established **centralized blocking** of devices by configuring rules and updating the rules on all systems

POKER AT-7 — Mentor: Prof. Purushottam Kar, CSE, IIT Kanpur — ESC101 [Oct '18 - Nov '18]
- Created an online single-player **POKER** game with the game logic built in vanilla **JavaScript**
- BOT behavior was based on the **probability** of different outcomes possible in the current game scenario

Relevant Courses

Operating System	Advanced Algorithms	Computer Networks
Machine Learning	Database Management	Compiler Design
Data Mining*	Linear Algebra and ODEs	Computer Organisation
Data Structures and Algorithms	Software Ops. and Development	Theory of Computation
Principles of Programming Languages*	Probability	Discrete Mathematics

*: Ongoing

Technical Skills

- **Languages :** C, C++ , Python, C#, Javascript, NodeJS, Bash Scripting, x86 assembly, HTML5
- **Tools :** Github, LaTeX, AWS, Pandas, Matplotlib, Numpy, SQL, MongoDB, Unity Engine

Positions of Responsibility

Senator — Students' Gymkhana, IIT Kanpur [Aug '18 - Mar '19]
- A representative **elected by 900 students** to act as their voice and put forth their interests in **Students' Senate**.